Statistical inference and superstatistics for power grid modeling

Alessandro Lonardi (Queen Mary)

w/

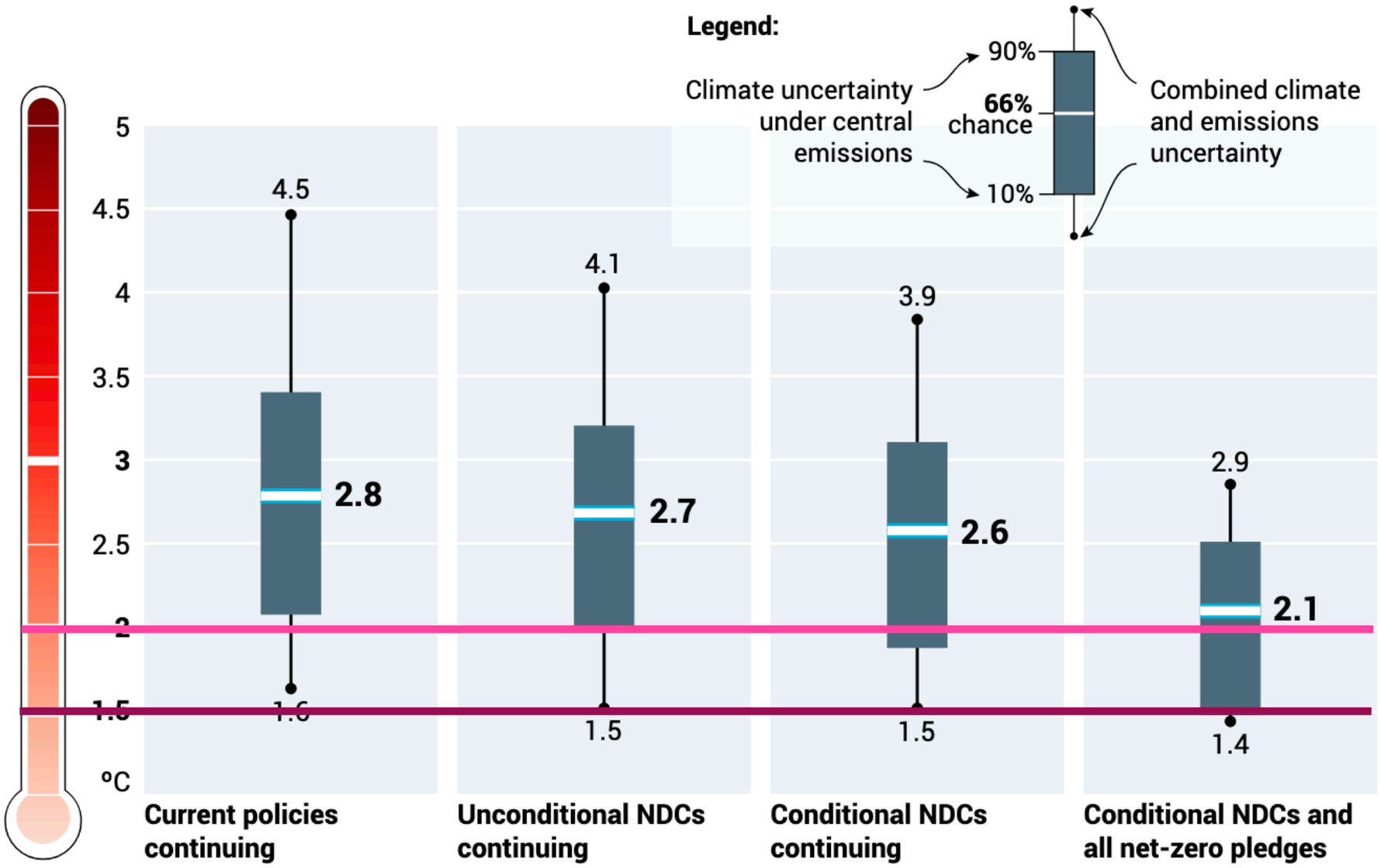
Benjamin Schäfer,
Jacques M. Maritz,
Leonardo Rydin Gorjão,
Christian Beck

a.lonardi@qmul.ac.uk

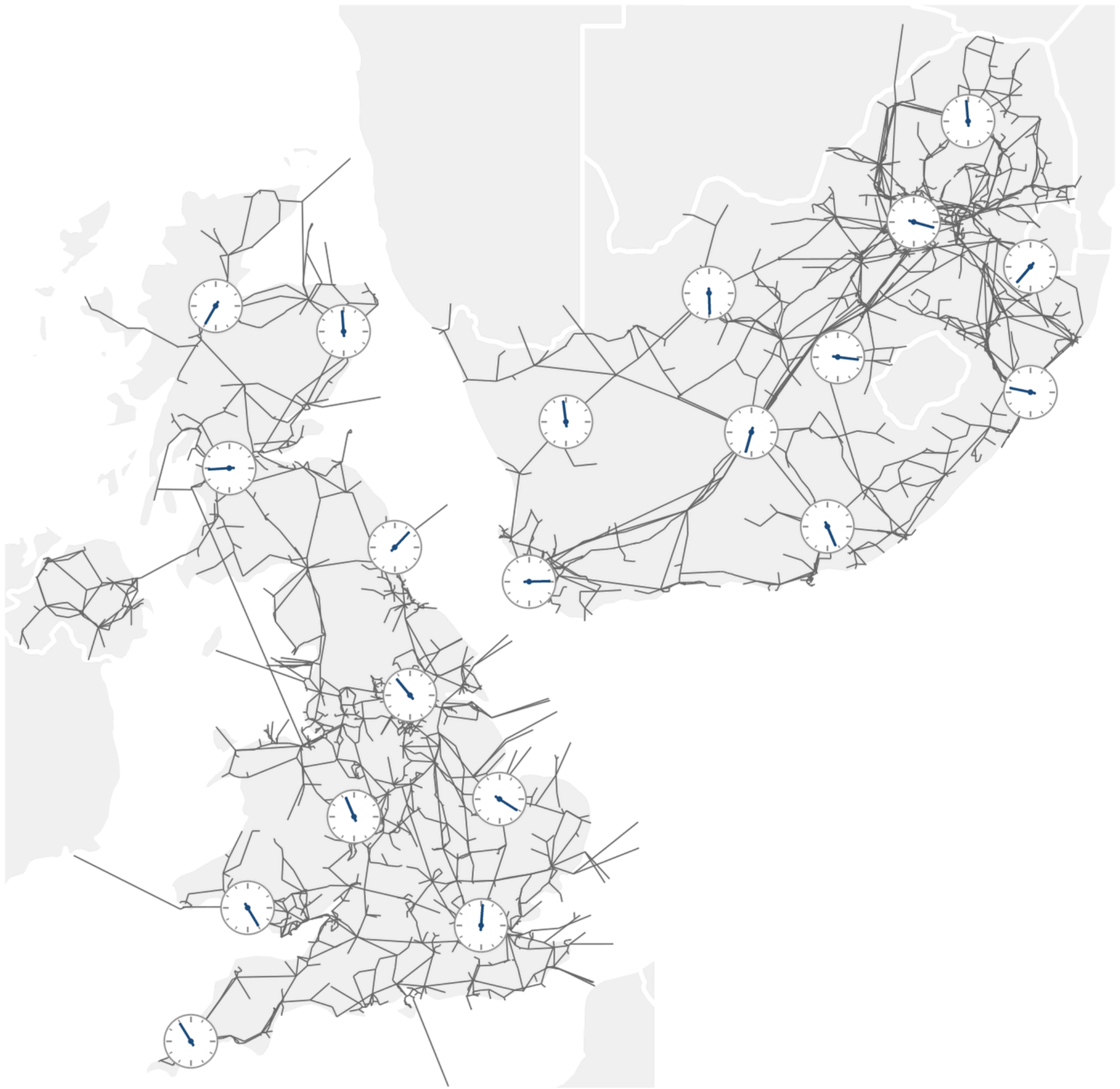
aleable.github.io

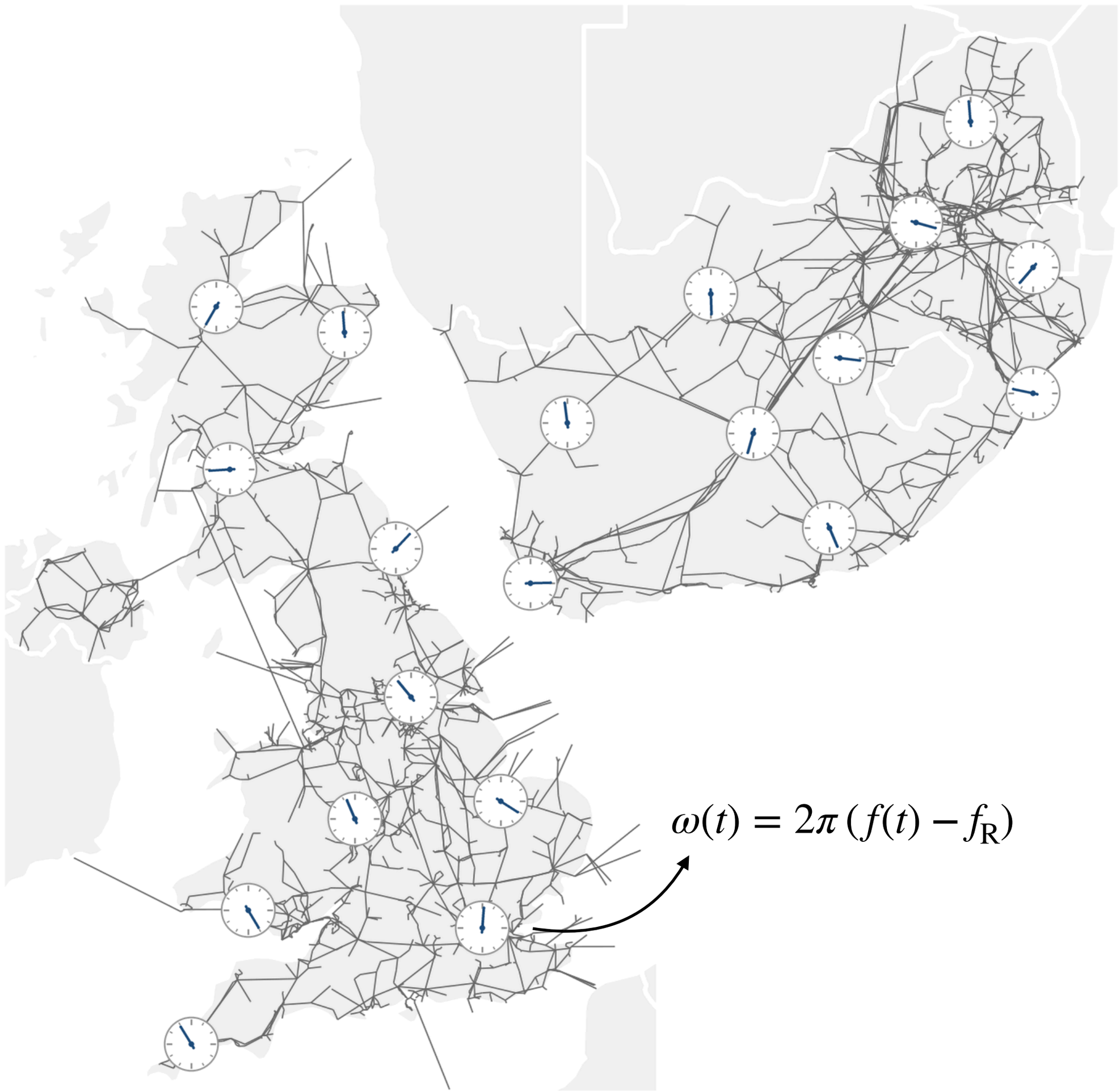
COMPLEX NETWORKS

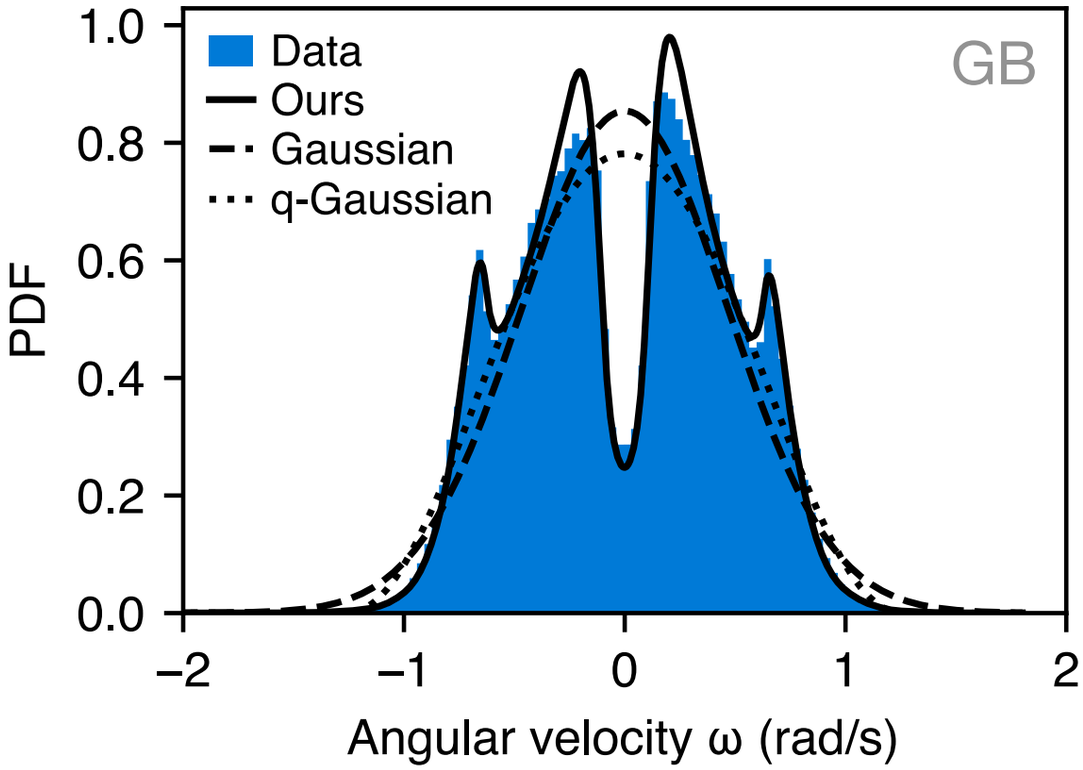
The Paris Agreement is failing

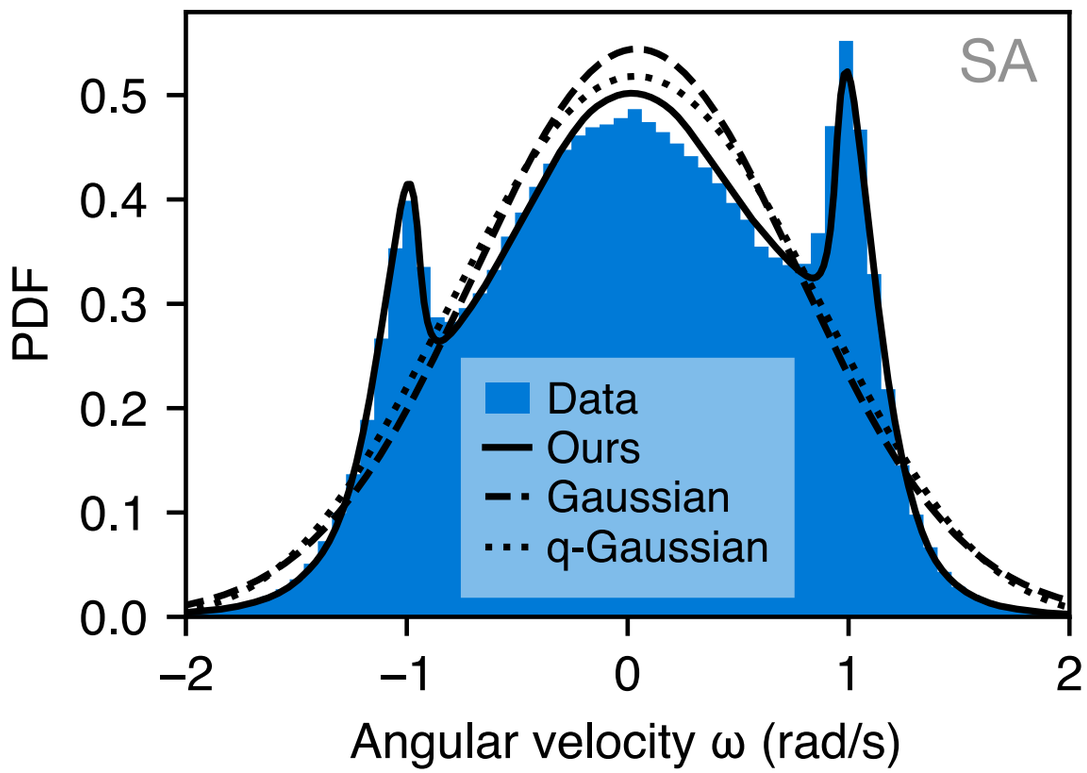


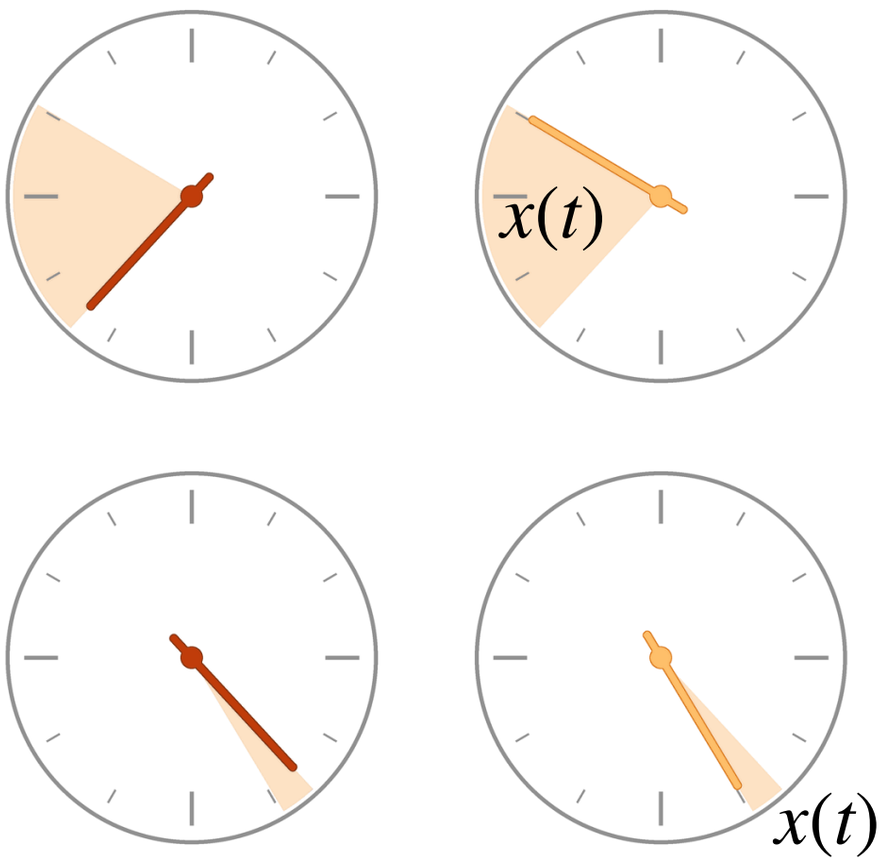
UNEP Emissions Gap Report, 13 October 2026

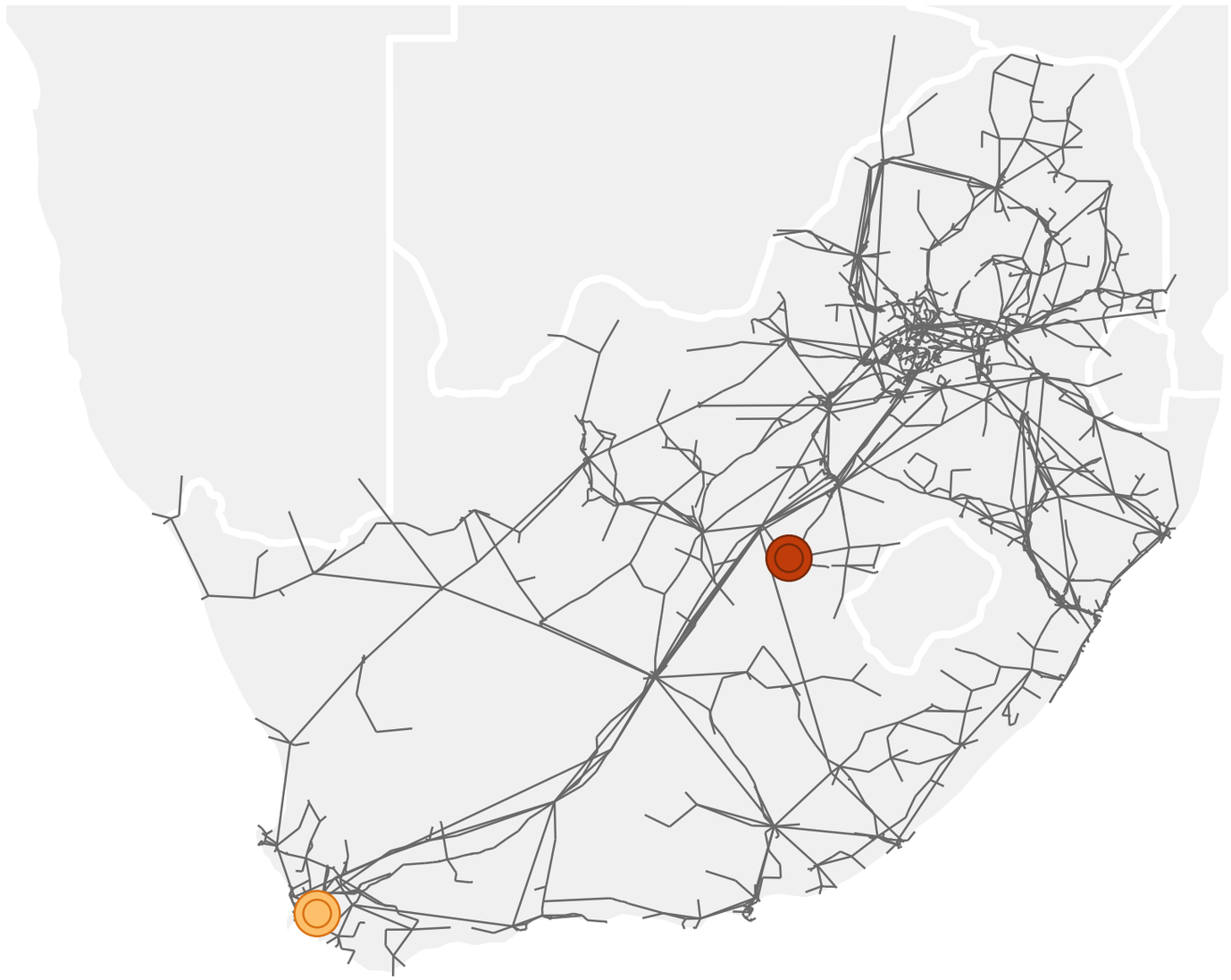


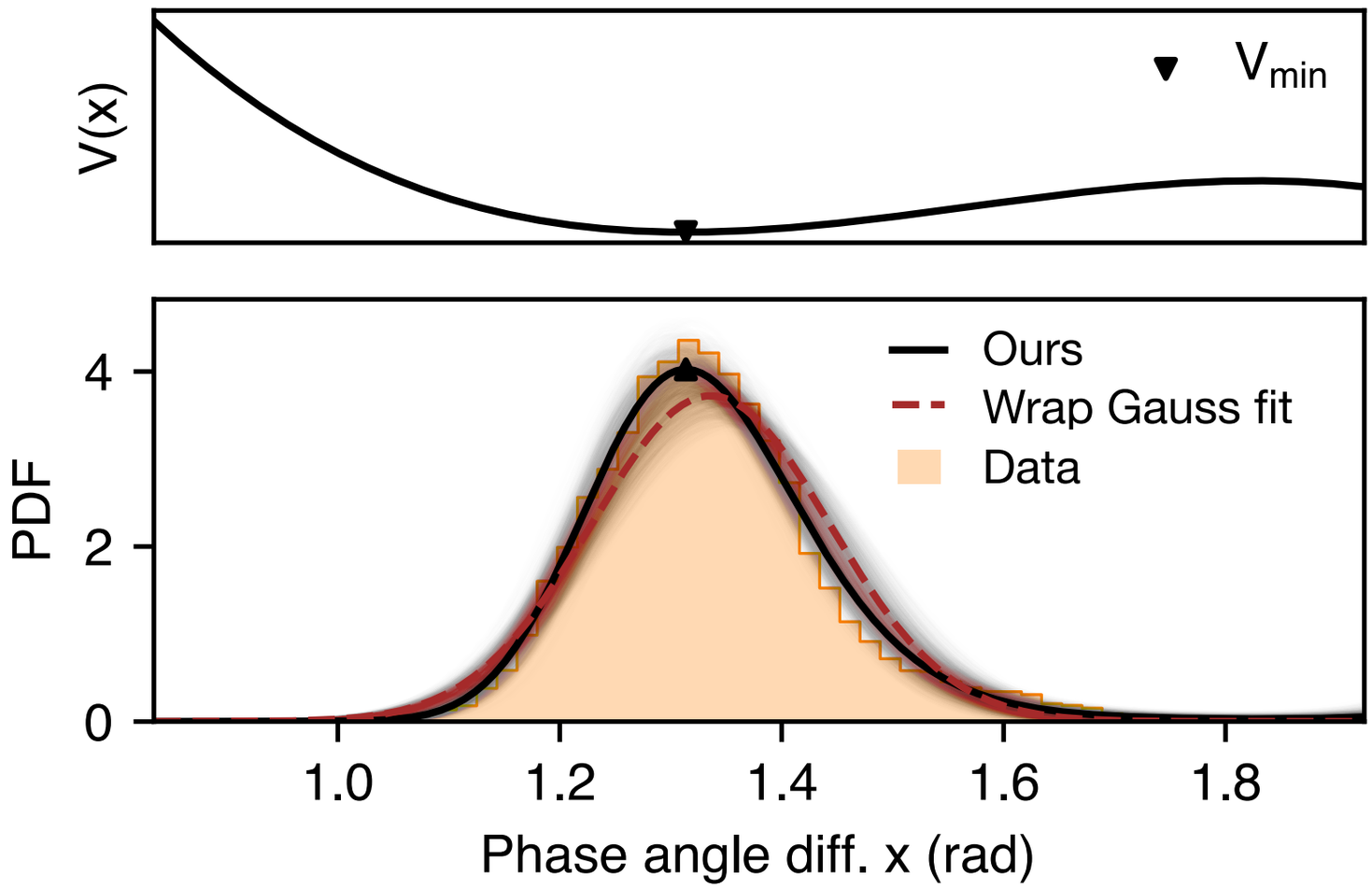

$\omega(t) = 2\pi\,(f(t) - f_{\mathrm{R}})$


GB
Data
Ours
Gaussian
q-Gaussian
PDF
1.0
0.8
0.6
0.4
0.2
0.0
−2
−1
0
1
2
Angular velocity ω (rad/s)


SA
Data
Ours
Gaussian
q-Gaussian
PDF
0.5
0.4
0.3
0.2
0.1
0.0
−2
−1
0
1
2
Angular velocity ω (rad/s)


$x(t)$
$x(t)$


Bloemfontein
Stellenbosch


V(x)
V_{min}
PDF
4
2
0
Ours
Wrap Gauss fit
Data
1.0
1.2
1.4
1.6
1.8
Phase angle diff. x (rad)

- AL, Maritz, Rydin Gorjão, Beck, PRX Energy
- AL, Schäfer, Beck, arXiv:2607.28232

Papers, codes, data, slides, …:

a.lonardi@qmul.ac.uk

aleable.github.io